Truong Buu Phan

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Google Scholar - LinkedIn

RESEARCH INTERESTS

I work at the intersection of information theory, tokenization, and machine learning. My research develops information-theoretic and statistical methods for neural compression and tokenization, targeting efficient video/image compression, language modeling, and scalable AI systems. Overall, I enjoy developing new foundational tools for the next generation of efficient, reliable, and trustworthy AI systems.

Ressearch Topics: LLMs, tokenization, information theory.

ACADEMIC BACKGROUND

Ph.D. Electrical and Computer Engineering — 2022 - Oct 2026
University of Toronto, Canada
- Advisor: Prof. Ashish Khisti.
- Research topic: sampling, compression and representation learning.
- Thesis: Probabilistic and Coding Techniques for One-Shot Compression and Language Modelling.

MASc. Electrical and Computer Engineering — 2017-2019
University of Waterloo, Canada
- Advisor: Prof. Krzysztof Czarnecki.
- Research in Bayesian deep learning and out-of-distribution detection.

BEng. Electrical Engineering — 2012-2016
Vietnam National University, Vietnam.

RESEARCH EXPERIENCE

Microsoft Research, USA — June 2026 - Aug 2026
Research Intern.
Host: Dr. David Alvarez-Melis.
Project: Theory of Scaling Laws.

Qualcomm AI Research, Canada — May 2025 - Aug 2025
AI Research Intern.
Host: Dr. Roland Memisevic.
Project: Length Generalization in Hybrid Recurrent-Transformer.

Meta AI (FAIR), USA — Mar 2024 - Oct 2024
AI Research Intern (Language Model and Probabilistic Reasoning).
Host: Dr. Karen Ullrich.
Project: Probabitistic reasoning, tokenization, LLMs.

LG Electronics AI Lab, Canada — May 2021 - May 2022
Research Engineer
Project: Automated check-out and neural symbolic AI.

Algolux *(acquired by Torc Robotics in 2023), Canada* — Oct 2019 - May 2021
Research Scientist
Collaborators: Prof. Felix Heide and Dr. Fahim Mannan.
Projects: Adversarial attack and robust vision for autonomous driving.

SELECTED PUBLICATIONS

For full publications, see my Google Scholar.

Buu Phan, Gergely Flamich, Ashish Khisti and Shahab Asoodeh. "Multi-Marginal Couplings for Metropolis-Hastings " (**Under Review**)
https://arxiv.org/abs/2605.12807.

Buu Phan, Ashish Khisti and Karen Ullrich. "Cross-Tokenizer Likelihood Scoring Algorithms for Language Model Distillation" (**International Conference on Learning Representations 2026**)
https://arxiv.org/pdf/2512.14954.

Joseph Rowan, **Buu Phan**, Ashish Khisti. "List-Level Distribution Coupling with Applications to Speculative Decoding and Lossy Compression" (**Conference on Neural Information Processing Systems 2025**).
https://arxiv.org/abs/2506.05632.

Buu Phan, Reza Ebrahimi, Sanjay Haresh, Roland Memisevic. "Delayed Attention Training Improves Length Generalization in Transformer–RNN Hybrids" (**What Can('t) Transformers Do? Workshop @ Conference on Neural Information Processing Systems 2025**).
https://arxiv.org/abs/2510.00258

Buu Phan, Brandon Amos, Itai Gat, Marton Havasi, Matthew Muckley and Karen Ullrich. "Exact Byte-Level Probabilities from Tokenized Language Models for FIM-Tasks and Model Ensembles" (**International Conference on Learning Representations 2025**). https://arxiv.org/abs/2410.09303.

Buu Phan, Ashish Khisti. "Channel Simulation and Distributed Compression with Ensemble Rejection Sampling" (**Conference on Neural Information Processing Systems 2025**).
https://www.arxiv.org/abs/2510.05552

Buu Phan*, Ashish Khisti*, and Christos Louizos. "Importance matching lemma for lossy compression with side information." **(International Conference on Artificial Intelligence and Statistics 2024)**.
https://arxiv.org/abs/2401.02609

Buu Phan, Ashish J Khisti "On List Decoding with Importance Sampling". (**International Symposium on Information Theory 2025**).
https://ieeexplore.ieee.org/abstract/document/11195600

Salehkalaibar Sadaf*, **Buu Phan***, Jun Chen, Wei Yu, and Ashish Khisti: *On the Choice of Perception Loss Function for Learned Video Compression.* (**Conference on Neural Information Processing Systems 2023**).
Spotlight in Neural Compression Workshop, International Conference on Machine Learning 2023.
https://arxiv.org/abs/2305.19301

Sadaf Salehkalaibar, **Buu Phan**, João Atz Dick, Ashish J Khisti, Jun Chen, Wei Yu "Perception Loss Function Adaptive to Rate for Learned Video Compression". In Machine Learning Compression Workshop, **NeurIPS Workshop** 2024.

Sadaf Salehkalaibar, **Buu Phan**, Ashish Khisti, and Wei Yu. "Rate-Distortion-Perception

Tradeoff Based on the Conditional Perception Measure." In 2023 **Biennial Symposium on Communications (BSC)**, IEEE, 2023.
https://ieeexplore.ieee.org/document/10201822

Buu Phan, Fahim Mannan, and Felix Heide: *Adversarial imaging pipelines.* In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, **(CVPR 2021)**. https://arxiv.org/abs/2102.03728

Buu Phan, Samin Khan, Rick Salay, and Krzysztof Czarnecki: *Bayesian uncertainty quantification with synthetic data.* In Computer Safety, Reliability, and Security: **(WAISE-SAFECOMP)** 2019 Workshops **(Best paper award).**
https://link.springer.com/chapter/10.1007/978-3-030-26250-1_31.

Nicolas Scheiner, Florian Kraus, Fangyin Wei, **Buu Phan**, Fahim Mannan, et al. "Seeing Around Street Corners: Non-Line-of-Sight Detection and Tracking In-the-Wild Using Doppler Radar." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. **(CVPR)** 2020.
https://arxiv.org/abs/1912.06613

OTHER PUBLICATIONS

Samin Khan, **Buu Phan**, Rick Salay, and Krzysztof Czarnecki. "ProcSy: Procedural Synthetic Dataset Generation Towards Influence Factor Studies Of Semantic Segmentation Networks." **CVPR Workshops** 2019.

Sachin Vernekar, Ashish Gaurav, Taylor Denouden, **Buu Phan**, Vahdat Abdelzad, Rick Salay, and Krzysztof Czarnecki. "Analysis of confident-classifiers for out-of-distribution detection." **ICLR SafeML Workshop** 2019. https://arxiv.org/abs/1904.12220

Buu Phan, Rick Salay, Krzysztof Czarnecki, Vahdat Abdelzad, Taylor Denouden, Sachin Vernekar, "Calibrating Uncertainties in Object Localization Task", **NeurIPS Bayesian Deep Learning Workshop**, 2018. https://arxiv.org/abs/1811.11210

Ian Colwell, **Buu Phan**, Shahwar Saleem, Rick Salay, and Krzysztof Czarnecki. "An automated vehicle safety concept based on runtime restriction of the operational design domain." In **2018 IEEE Intelligent Vehicles Symposium (IV)**.

Denouden Taylor, Rick Salay, Krzysztof Czarnecki, Vahdat Abdelzad, **Buu Phan**, and Sachin Vernekar. "Improving reconstruction autoencoder out-of-distribution detection with mahalanobis distance", preprint 2018. https://arxiv.org/abs/1812.02765

AWARDS

Shahid U. H. Qureshi Memorial Scholarship — 2026
Award for top students in ECE Department.

Ontario Graduate Scholarship — 2023
Award for top students in Ontario, Canada.

Best paper award — 2019
Received at the Workshop on Artificial Intelligence Safety Engineering for the paper "Bayesian uncertainty quantification with synthetic data."

Toyota Canada Graduate Scholarship — 2018
Scholarship for graduate students working in AI Safety.

Faculty of Engineering Awards, University of Waterloo — 2018, 2019

Scholarship for top-performing graduate students.

International Master's Student Awards, University of Waterloo 2017-2019
Scholarship for international graduate students.

Undergraduate Valedictorian, Vietnam National University (IU Campus) 2016

ACADEMIC SERVICES

- Journal Reviewer - TMLR, IEEE.
- Conference Reviewer - AISTATS/CVPR/ICLR/NeurIPS.

SOFTWARE SKILLS

Programming: PyTorch, HuggingFace, Tensorflow, Matlab, PyThon, C/C++.
System: Unix, MacOS .
Tools: Git, Slurm, Docker, WandB.